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MALWARE DETECTION SYSTEM USING SIGNATURE MATCHING TECHNIQUES FOR SECURE FILE SCANNING AND THREAT IDENTIFICATION

PRESENTED BY

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INTRODUCTION



- Malware is malicious software designed to damage computer systems, steal confidential information, or disrupt normal system operations.
- With the rapid growth of the internet, cyber threats have increased significantly, making malware detection an essential part of computer security.
- Signature matching is a widely used malware detection technique that identifies malicious files by comparing their unique signatures with a database of known malware signatures.
- Java is a platform-independent, object-oriented programming language that provides secure file handling and efficient processing, making it suitable for developing security applications.
- Early malware detection helps protect sensitive data, improves system security, reduces the risk of cyberattacks, and ensures safe file management.



PROBLEM STATEMENT



- The increasing number of malware attacks poses a serious threat to computer systems, leading to data loss, privacy breaches, and system failures.
- Many existing antivirus solutions are expensive, resource-intensive, and may require frequent internet connectivity for signature updates.
- Detecting malicious files quickly and accurately is challenging, especially for users with limited access to advanced cybersecurity tools.
- There is a need for a lightweight and efficient application that can securely scan files and identify known malware using signature matching techniques.
- The project aims to provide a Java-based malware detection system that helps users detect known threats, improve file security, and generate reliable scan reports.



OBJECTIVES



- To Develop a Java application for malware detection.
- To Scan files stored on the user's system.
- To Generate file signatures using hashing techniques.
- To Compare generated signatures with a malware signature database.
- To Detect and identify malicious files accurately.
- To Display detailed scan results to the user.
- To Maintain a database of malware signatures for future updates.
- To improve system security through early malware detection.



PROPOSED SYSTEM

- The proposed system is a lightweight Java application that scans files using signature matching techniques. It calculates a unique signature (hash value) for each file and compares it against stored malware signatures.

Features

- Secure file scanning.
- Malware signature database.
- Fast signature matching.
- Threat identification.
- User-friendly interface.
- Scan history generation.
- Report generation.



Module 1: File Scanning



- Allows users to select individual files or folders for scanning.
- Reads the contents of the selected files securely.
- Supports scanning of different file formats.
- Checks files before analysis begins.
- Processes multiple files efficiently.
- Reduces the chances of missing suspicious files.
- Sends the scanned files to the signature matching module.



Module 2: Signature Matching



- Generates a unique hash (SHA-256/MD5) for each scanned file.
- Loads the malware signature database.
- Compares generated signatures with stored malware signatures.
- Identifies files that match known malware signatures.
- Detects known malware quickly and accurately.
- Minimizes false detection through exact signature comparison.
- Sends the detection results to the threat detection module.



Module 3: Threat Detection



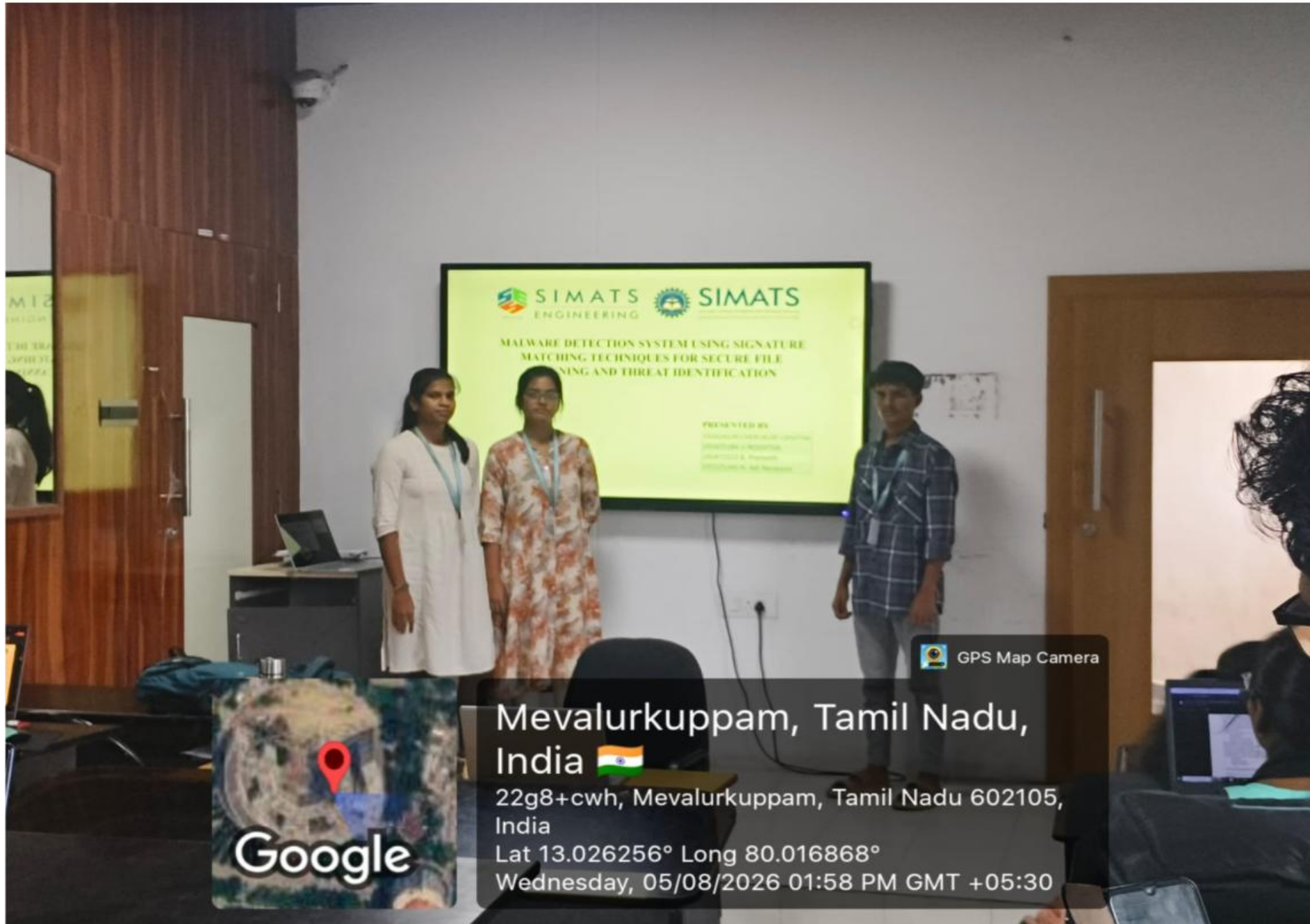
- Analyzes the results received from the signature matching module.
- Determines whether the scanned file is safe or infected.
- Classifies detected files as malicious or non-malicious.
- Displays warning messages for infected files.
- Provides details about the detected malware.
- Alerts users to potential security risks.
- Passes the final detection status to the report generation module.



Module 4: Report Generation



- Generates a detailed report after every scan.
- Displays the scanned file name and location.
- Records the scan date and time.
- Shows whether the file is safe or infected.
- Includes malware details for detected threats.
- Saves scan history for future reference.
- Provides an easy-to-understand summary of the scan results.



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